Battery- and Health-Aware Drone Traffic Management: Research Scope

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Summary

Novel claim: MASooda closes the loop between every drone's live battery state and the multi-agent traffic manager. Who yields, who gets which altitude layer, who lands first and who absorbs the energy cost of avoiding a conflict are decided from each drone's energy margin, thermal margin and BMS status. The loop is also protected against drones that lie about their battery health.

The literature has the two halves but not the join. Battery prognostics for drones are mature, but they decide only for one vehicle: NASA's 2024 flight tests land the same aircraft when remaining flight time drops below 200 s [Ancel 2024]. Traffic management, meanwhile, sets priority from operation class and geometry, not battery state [Jover 2022]. The closest work uses battery state only as a deadline in offline vertiport arrival sequencing, without en-route conflicts [Kleinbekman 2018].

MASooda already has a physics-based health model, BMS fault codes, onboard diversion and a basic "unhealthy drone gets right of way" DTC rule. The publishable step is to make that rule continuous, push it into ADMM conflict resolution and STL safety monitoring, and defend it against falsified health reports.

Battery and health models

These are the physics behind every drone's health state, ticked every 0.1 s. Sections 1–5 are already implemented in MASooda's health model; section 6 (degradation across flights) is proposed.

1. Air density and power draw

Air density at the site comes from elevation h and ambient temperature T_amb. Hot air lifts less, so hover costs more:

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\rho = \frac{101325\,(1 - 2.25577\times10^{-5} h)^{5.25588}}{R_d\,(T_{amb} + 273.15)}
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For Hyderabad (542 m) this gives 1.074 kg/m³ at 35 °C and 1.050 kg/m³ at 42 °C. Momentum theory [Leishman 2006] gives the induced velocity for thrust T = m·g over rotor disc area A, at airspeed v:

$$v_h = \sqrt{\frac{T}{2\rho A}}, \qquad v_i(v) = \sqrt{-\frac{v^2}{2} + \sqrt{\frac{v^4}{4} + v_h^4}}$$

Mechanical power is induced plus parasitic plus climb power; electrical power adds propulsion losses, motor health and avionics:

$$P_{mech} = \frac{T\,v_i}{FM} + \tfrac{1}{2}\rho\,C_D A_f\,v^3 + m g\,v_{climb}, \qquad P_{elec} = \frac{P_{mech}}{\eta_{prop}\,h_{motor}} + P_{avionics}$$

2. Battery equivalent circuit and state of charge

The pack is an internal-resistance (Rint) circuit [Plett 2015]. Open-circuit voltage OCV follows a Li-ion curve of state of charge (SoC); resistance rises with ageing and cold:

$$R = R_{pack}\,\bigl(1 + 0.5\,(1 - SoH)\bigr)\,\bigl(1 + \alpha_R\,(25 - T_{pack})\bigr)_+$$

The current that delivers P_elec, terminal voltage and SoC update are:

$$I = \frac{OCV - \sqrt{OCV^2 - 4 R P_{elec}}}{2R}, \qquad V = OCV - I R, \qquad SoC_{t+\Delta t} = SoC_t - \frac{I\,\Delta t}{3600\,Q_{nom}\,SoH}$$

If OCV² < 4·R·P_elec, the pack cannot deliver the power, which is a BMS FAULT (voltage collapse). A Thevenin 1-RC branch would add voltage-sag transients under climb loads; it is an optional refinement.

3. Usable energy and energy to destination

Remaining usable energy counts down to SoC_min = 5 %:

$$E_{rem} = Q_{nom}\,SoH \int_{SoC_{min}}^{SoC} OCV(s)\,ds$$

The energy needed to reach a pad L metres away at cruise speed, with a 20 % margin and a 30 s landing, is:

$$E_{need}(L) = P_{elec}(v_{cr})\,\frac{L}{v_{cr}}\,(1 + m) + P_{elec}(0)\,t_{land}$$

Today this check runs every 1 s and triggers a diversion when E_rem < E_need(destination).

4. Temperature

Pack and motor temperatures follow first-order lumped models [Bernardi 1985, simplified], with cooling that improves with airspeed:

$$C_{pack}\,\frac{dT_{pack}}{dt} = I^2 R + q_{fault} - hA(v)\,(T_{pack} - T_{amb}), \qquad hA(v) = hA_0 + hA_1\sqrt{v}$$

5. BMS status

Each category (low charge, pack and motor temperature, cell imbalance, over-current, sensor fault) maps to NORMAL, WARNING, CRITICAL or FAULT, with hysteresis so the status does not flicker. For example, SoC below 30 % is WARNING and below 15 % is CRITICAL; a pack above 65 °C is CRITICAL.

6. Degradation across flights (proposed)

To study fleet-level effects such as which drones age fastest under a given traffic policy, add an Arrhenius cycle-ageing model [Wang 2011]. Capacity loss grows with throughput Ah and temperature:

$$Q_{loss}\,[\%] = B \exp\!\left(-\frac{E_a}{R\,T}\right) (Ah)^{z}, \qquad SoH = 1 - \frac{Q_{loss}}{100}$$

This lets the paper report battery wear per delivery as a cost of each traffic policy, not just energy per flight.

What research already exists

Each strand below solves part of the problem; none feeds live battery state of many drones into traffic decisions.

Theme	Work	What it does	What it lacks for our problem
Priority in conflict resolution	Jover et al. 2022	Priority from operation class and manoeuvrability in U-space tactical deconfliction; low-priority drones fly up to 8 % farther	Battery and health play no part in priority
Energy cost of deconfliction	Zongo & Wei 2026	Median energy overhead below 1.5 %, 95th percentile 3.84–5.3 %, tails to 44 % at high density; recommends a 4–5 % reserve	Overhead is estimated at mission start; live battery state does not steer manoeuvres
Battery-constrained sequencing	Kleinbekman, Mitici & Wei 2018	Battery state sets each eVTOL's latest arrival time in a MILP vertiport schedule	Terminal area only, no en-route conflicts; 80 aircraft took 5,333 s to solve
Energy-based flight planning	Liu & Sengupta 2017	Pre-flight plans that minimise energy for UTM	Pre-flight only; no live health or tactical decisions
Health-aware single-vehicle operation	Montazeri, Kulkarni & Fink 2024	Deep RL picks destinations and cruise altitude from inferred SoH (capacity, resistance)	Single aircraft; no traffic, separation or adversarial reports
Onboard battery prognostics	Saha et al. 2011, Kulkarni et al. 2018, Hogge et al. 2024	Remaining-flight-time prediction, flight-tested on electric UAVs and octocopters	Decisions stay onboard one vehicle

Theme	Work	What it does	What it lacks for our problem
Run-time assurance	Ancel et al. 2024	Over 100 flights; battery monitor recommends landing when remaining flight time falls below 200 s	Single vehicle; no coordination with other traffic
Prognostics-informed reconfiguration	Sharma & Atkins 2021, Darrah 2023	Switch battery packs or replan after faults using prognostics	One vehicle; no shared airspace or landing capacity
Formal safety for fleets	Pant et al. 2018	STL robustness for multi-drone separation and missions	No energy or thermal specifications
Resilient consensus	LeBlanc et al. 2013	W-MSR tolerates misbehaving neighbours	Applied to positions and states, never to health telemetry
Attacks on battery estimates	Lin, Li & Meng 2026	RL false-data injection steers cloud-BMS SoH estimates ($R^2 > 0.97$) while evading detectors	Batteries in general; no traffic-management consequence studied
Battery physics for vertical flight	Yang et al. 2021, Zhao et al. 2025	eVTOL batteries need more than EV batteries in every aspect; reliability review lists cybersecurity and multi-fault diagnosis as open	Component level, not operations

The nearest overlap is Kleinbekman 2018 (battery as an arrival deadline) and Montazeri 2024 (health-aware decisions). Both must be cited and contrasted explicitly in the paper.

Ongoing research and programmes

Active work in 2024–2026 is moving health monitoring into operations, but still one aircraft at a time.

Group / programme	Status (2024–2026)	Relevance to MASooda
NASA Langley, in-time safety for autonomous UAS	Run-time assurance flight tests with battery, propulsion and autopilot monitors [Ancel 2024]; octocopter battery prognostics [Hogge 2024]; the HERMES risk-mitigation system, presented January 2026 with battery prognostics as a keyword (NTRS)	Supplies exactly the signals our DTC would consume; its decisions are per vehicle
NASA IASMS concept for Advanced Air Mobility	Lists vehicle-health monitors and battery prognostics as in-time risk services that can trigger mitigation [Ellis 2021]	Concept asks for health-driven mitigation; the multi-vehicle policy is open
India National UTM Policy Framework (2021)	Lists real-time UAS health monitoring and contingency management as UTM services [MoCA 2021]	Direct policy hook for a health-aware DTC in the Indian context
FAA UTM concept v2.0	Endurance stays the operator's responsibility [FAA 2020]	Shows that today's traffic layer does not own energy
George Washington Univ. (Peng Wei's group)	Energy overhead of deconfliction published 2026 [Zongo & Wei 2026]; earlier battery-constrained arrival sequencing [Kleinbekman 2018]	Most active competitor line; our angle (live health in the loop + adversarial reports) is distinct
EPFL / NASA Ames (Fink, Kulkarni)	Health-aware RL mission planning [Montazeri 2024]	Single-vehicle; fleet operation named as future work

Group / programme	Status (2024–2026)	Relevance to MASooda
Battery cybersecurity	Stealthy RL attacks on SoH estimation, 2026 [Lin 2026]	Motivates our health-report trust layer

The trend is clear: monitoring is maturing, and the open question every programme points to is the multi-vehicle policy that uses it. That is where MASooda sits.

The gap

No work we found combines live, multi-vehicle battery state with tactical traffic decisions and defends that input against falsification. Six specific gaps follow (refined from the team's 7 Nov review):

#	Gap	Evidence
G1	The traffic manager is health-blind: priority, layers and deconfliction weights ignore SoC, SoH and temperature	Jover 2022; FAA 2020
G2	Prognostics stop at one vehicle, without knowledge of surrounding traffic or shared landing pads	Ancel 2024; Hogge 2024; Montazeri 2024
G3	The energy cost of deconfliction is measured, not managed: a low-reserve drone can still be the one forced to manoeuvre	Zongo & Wei 2026
G4	Health telemetry is unprotected once it drives priority: a drone could fake CRITICAL to jump the queue, or hide a real fault	Lin 2026 (attacks exist on estimates); no traffic-level study found
G5	Safety specifications ignore energy and heat: STL covers separation, not reserve or temperature	Pant 2018
G6	No traffic-management study evaluates heat-wave conditions (40–44 °C) and their effect on derating, diversions and throughput	Policy asks for health monitoring (MoCA 2021); no study found

Caveat: G4 and G6 are absences in our searches, not proven absences. Re-search before submission and read the full texts of Kleinbekman 2018, Montazeri 2024 and Zongo & Wei 2026, whose details here come partly from abstracts.

Link to MASooda

The health foundation is built; the traffic-side uses of it are mostly not. Status as of the p2_week7 merge:

Capability	Status	Where
Power, battery circuit, temperature and BMS model (sections 1–5 above)	Built	simulation/drone_health.py
Fault injection: capacity loss, cell failure, overheating, motor degradation, GNSS degradation	Built	drone_health.py, scenario presets
Onboard response: low-reserve diversion, forced landing, overheat speed limit	Built	simulation/route_engine.py
DTC right of way for CRITICAL/FAULT drones, lower SoC first, landing pad reserved, no energy-costly climb	Built, binary rule	route_engine.py
Fleet recall below a SoC threshold	Built	simulation/dtc_console.py
Continuous health score in DTC priority and layer choice	Not yet	overseer.py
Energy-weighted ADMM (who absorbs the manoeuvre)	Not yet	conflict_resolution/
Energy and thermal STL specifications	Not yet	verification/
Trust checks on health reports	Not yet (W-MSR covers positions only)	resilience/
Energy-aware altitude layers	Designed (Altitude Layer Design doc)	overseer.py, config
Health model in the main experiment engine (engine.py)	Not yet: health lives in the live-demo route engine only	simulation/engine.py

The last row matters for the paper: the Week 6 experiments run on engine.py, so health must reach that engine, or the experiments move to the route engine, before results can be reported. The altitude-layer separation bug from the 9 Nov review must be fixed first in either case.

Proposed contributions

Five contributions, ordered by value to the paper. C1–C3 form the core claim; C4 is the strongest differentiator; C5 grounds it in Hyderabad conditions.

C1. Continuous health score for DTC priority (closes G1, G2)

Replace the binary CRITICAL rule with margins. For drone i, the energy and thermal margins are:

```
m_{E,i} = \frac{E_{rem,i} - E_{need,i}(\text{nearest pad})}{E_{need,i}
(\text{nearest pad})}, \qquad m_{T,i} = \frac{T_{crit} - T_{pack,i}}{T_{crit}
- T_{amb}}
```

The health urgency combines them with the BMS level b_i (0 = NORMAL … 3 = FAULT):

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u_i = w_E\,\sigma\!\left(\frac{\bar m_E - m_{E,i}}{s_E}\right) +
w_T\,\sigma\!\left(\frac{\bar m_T - m_{T,i}}{s_T}\right) + w_B\,\frac{b_i}
{3}, \qquad \pi_i = \min\bigl(\pi_{type,i},\; \lfloor 3(1 - u_i)
\rfloor\bigr)
```

σ is the logistic function, so urgency rises smoothly as margins shrink below their thresholds. Effective priority π_i takes the more urgent of mission type and health. The DTC then sequences landings by π_i and, within a pad, by the latest feasible arrival time implied by E_rem (the online, en-route version of Kleinbekman 2018).

C2. Energy-fair ADMM conflict resolution (closes G3)

Today's ADMM chooses velocities close to each drone's desired velocity, weighted by type priority. Make the weight grow as energy margin shrinks, so drones with reserve absorb the manoeuvre:

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\min_{v_1..v_N} \sum_i w_i\,\lVert v_i - v_i^{des}\rVert^2 \;\text{s.t.
separation}, \qquad w_i = w_{type,i}\left(1 + \frac{\kappa}{\max(m_{E,i},
\varepsilon)}\right)
```

An optional hard constraint caps the extra energy any drone can be asked to spend, where ΔP_i is the power increase of the new velocity over the manoeuvre time τ:

$$\Delta E_i = \Delta P_i(v_i)\,\tau \le \beta\, m_{E,i}\, E_{need,i}$$

The paper reports the energy-overhead distribution (median, 95th percentile, maximum) with and without this, directly comparable to Zongo & Wei 2026.

C3. Energy and thermal STL shield (closes G5)

Add two specifications beside separation, each with a robustness value:

$$\varphi_E = \mathbf{G}\,(E_{rem} - E_{need}(\text{nearest pad}) - E_{res} \ge 0), \qquad \rho_E(t) = \min_i\bigl(E_{rem,i} - E_{need,i} - E_{res}\bigr)$$

$$\varphi_T = \mathbf{G}\,(T_{pack} \le T_{max}), \qquad \rho_T(t) = \min_i\, (T_{max} - T_{pack,i})$$

Normalising each by its scale gives one fleet safety robustness, ρ = min(ρ_sep/d_min, ρ_E/E_res, ρ_T/ΔT). The shield acts before violation using a time-to-violation estimate, where Ė_need is how fast the energy to the nearest pad changes as the drone flies:

$$t^{*}_{E,i} = \frac{E_{rem,i} - E_{need,i} - E_{res}}{P_{elec,i} - \dot E_{need,i}} \quad\Rightarrow\quad \text{DTC divert or re-sequence if } t^{*}_{E,i} < \tau_{act}$$

C4. Trusted health telemetry (closes G4)

Once health buys priority, a drone has a reason to lie. Two attacks matter: faking CRITICAL to jump the queue, and hiding a real fault. The DTC can check each report against physics it observes independently: tracked speed and climb imply a power draw, and the reported SoC must fall accordingly.

$$r_i(t) = \Bigl|\,\Delta SoC_i^{rep} - \frac{\hat P_{elec}(v_i^{obs}, \dot h_i^{obs})\,\Delta t}{3600\,V\,Q_{nom}\,SoH_i}\Bigr|, \qquad S_i(t) = \max\bigl(0,\; S_i(t-1) + r_i(t) - k\bigr)$$

A CUSUM statistic S_i above threshold h flags the drone. A trust score $\tau_i \in [0, 1]$ then blends reported and type-based priority, so an untrusted drone cannot gain right of way:

$$\pi_i^{eff} = \tau_i\,\pi_i^{health} + (1 - \tau_i)\,\pi_{type,i}$$

Peer reports of a neighbour's observed motion can be fused with W-MSR, reusing MASooda's existing resilience layer. The metrics are time to detection, false accusations and priority gained by the liar.

C5. Energy-aware layers and hot-climate operations (closes G6)

Use the 7-layer altitude design. Each drone's layer minimises climb plus cruise energy, with wind rising with height by a power law ($\alpha \approx 0.3$ assumed for urban terrain):

$$\ell_i^{*} = \arg\min_{\ell}\left[\frac{m_i g\,(h_\ell - h_0)}{\eta} + P_{elec}\bigl(v_{air}(h_\ell)\bigr)\frac{L_i}{v_{g,i}}\right], \qquad w(h) = w_{ref}\left(\frac{h}{h_{ref}}\right)^{\alpha}$$

The choice is subject to layer capacity and direction rules. Low-margin drones are restricted to the lowest feasible layer. All of this runs at 35 °C and at a 42 °C heat wave, where hotter air and worse pack cooling shrink margins across the fleet.

Applications

Setting	What health-aware traffic management changes
Medical delivery (blood, medicines, AEDs to a hospital pad)	A low-battery delivery drone is not forced to yield to routine traffic and land off-site with its payload; the hospital pad is sequenced by urgency and energy deadline
Campus and city parcel delivery fleets	Manoeuvre energy shifts to drones with reserve, cutting diversions and forced landings; battery wear per delivery becomes a managed cost (section 6 model)
Heat waves in Indian cities	The DTC tightens reserves and restricts layers when 40–44 °C air and hot packs shrink margins fleet-wide
Disaster response	Fleets recalled or re-sequenced by remaining energy when charging infrastructure is limited

Setting	What health-aware traffic management changes
eVTOL vertiports	Arrival sequencing uses live energy deadlines en route, not only at the terminal
Regulators and UTM service providers	A concrete design for the real-time health-monitoring and contingency services named in India's UTM policy framework
Security	Health-report trust checks stop a compromised or dishonest drone from gaming priority

Experiment and evaluation plan

The headline experiment compares three DTC policies on the IITH map under identical seeds.

Policies:

1. **Health-blind:** priority from mission type only (today's main engine).
2. **Binary rule:** CRITICAL/FAULT drones get right of way (today's live engine).
3. **Health-aware:** C1 + C2 + C3, with C4 switched on in the attack scenarios.

Factors:

Factor	Levels
Ambient temperature	35 °C (pre-monsoon), 42 °C (heat wave)
Fault scenario	None; capacity loss; cell failure; overheating
Health-report attack	None; fake CRITICAL to gain priority; hidden real fault
Fleet size	30, 60, 90 drones
Seeds	10 per cell

Metrics:

Metric	Definition	Target
Energy-exhaustion events	Drones reaching SoC_min airborne	0
Diversions and forced landings	Battery-driven events per 100 flights	Lower than both baselines
Energy-overhead distribution	Median, 95th percentile and maximum extra energy from deconfliction	Narrower tail than health-blind; compare with Zongo & Wei 2026
Emergency delay	Delay of emergency missions versus nominal	No worse than health-blind
Throughput	Arrivals per minute	Within 5 % of health-blind
Fleet safety robustness	Minimum of normalised ρ_{sep}, ρ_E, ρ_T	Positive throughout
Fairness	Jain's index of delay and of energy cost across drones	Higher than health-blind
Attack resilience	Time to detection, false accusations, priority gained by the liar	Fast detection, no false accusations

Ablations: C1 alone, C1 + C2, C1 + C2 + C3, each with and without C4, to show what each piece contributes.

Threats to validity: the battery model is a documented equivalent circuit, not calibrated to a specific cell; corridors are traced from a schematic map and OpenStreetMap; and the ADMM and DTC run in simulation, not on flight hardware. Each is stated in the paper.

Candidate venues: IEEE/AIAA Digital Avionics Systems Conference (DASC), where both Kleinbekman 2018 and Ancel 2024 appeared; IEEE ICUAS; and IEEE Transactions on Intelligent Transportation Systems for a full journal version.

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